DIABETIC RETINOPATHY CLASSIFICATION USING DEEP LEARNING AND ORDINAL REGRESSION.

Final Thesis Evaluation

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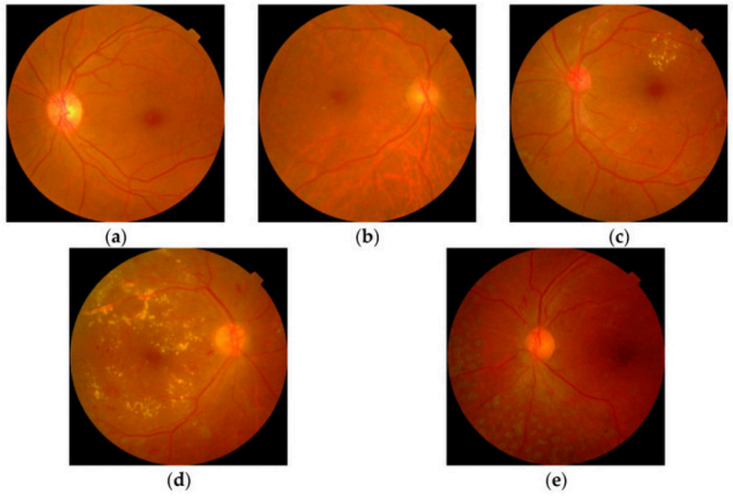
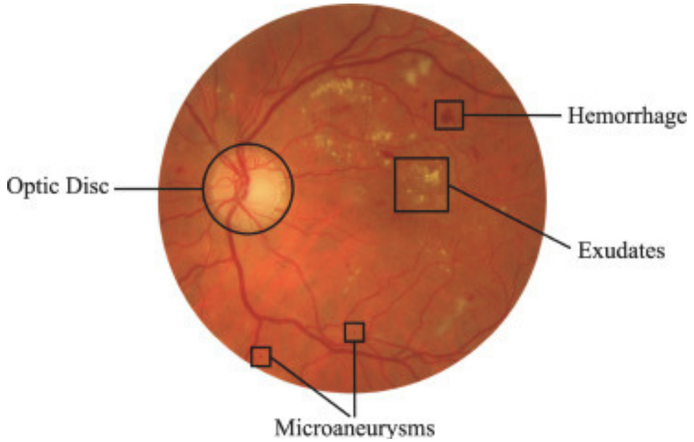
BACKGROUND & CLINICAL MOTIVATION

- Diabetic Retinopathy (DR) is a leading cause of preventable blindness worldwide.
- The ICDR severity scale ranges from Grade 0 (No DR) to Grade 4 (Proliferative DR).
- **The Problem**: Manual grading is time-consuming, subjective, and difficult to scale in regions with high diabetes prevalence.





THE GAP & RESEARCH

The Gap:

Most deep learning approaches treat DR grading as a nominal multi-class problem, ignoring the inherent ordinal nature of disease severity (Grade 0 < 1 < 2 < 3 < 4).

Objectives:

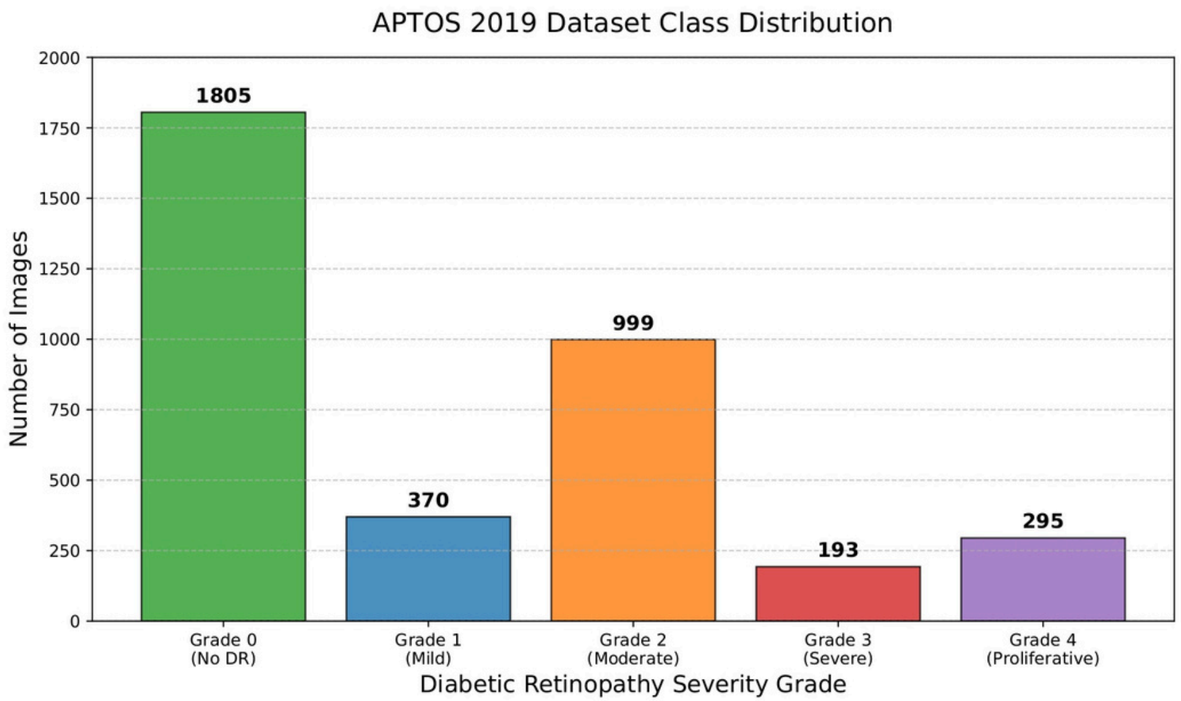
- Objective 1: Formulate DR grading using Ordinal Regression to penalize severe misclassifications clinically.
- Objective 2: Compare a CNN (EfficientNetV2-S), a Transformer (Swin-V2-Tiny), and a Foundation Model (RETFound).
- Objective 3: Build a heterogeneous late-fusion ensemble to capture complementary features.

DATASET & CHALLENGES

Source: Aravind Eye Hospital / Kaggle (APTOS 2019 Blindness Detection).

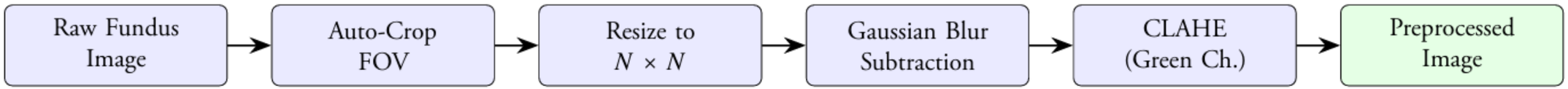
Dataset: APTOS 2019 Blindness Detection (3,662 images).

Challenges: Severe class imbalance, imaging variability (different resolutions, lighting), and subjective label noise.



PREPROCESSING PIPELINE

Consistent preprocessing is vital to standardize imaging variations.



Ben Graham Pipeline Steps:

1. Automated Field-of-View Cropping.
2. Resolution Standardisation (512 X 512 or 224 X 224).
3. Gaussian Blur Subtraction (removes illumination artifacts).
4. CLAHE on the Green Channel (enhances contrast of red lesions)

METHODOLOGY – ORDINAL REGRESSION FRAMEWORK

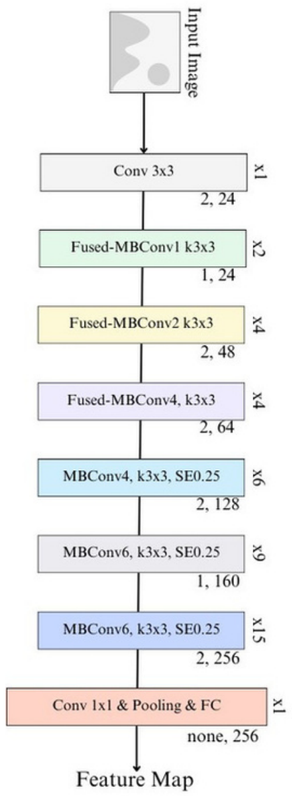
- **CORN (Conditional Ordinal Regression)**: Transforms the 5-class problem into 4 binary tasks ("Is severity > rank k?").
- Guarantees monotonically non-increasing cumulative probabilities without restricting model capacity.
- **Hybrid Loss Function**: Combines CORN Loss (for rank consistency) with Ordinal Focal Loss (to tackle class imbalance).
- This directly aligns with the Quadratic Weighted Kappa (QWK) evaluation metric.

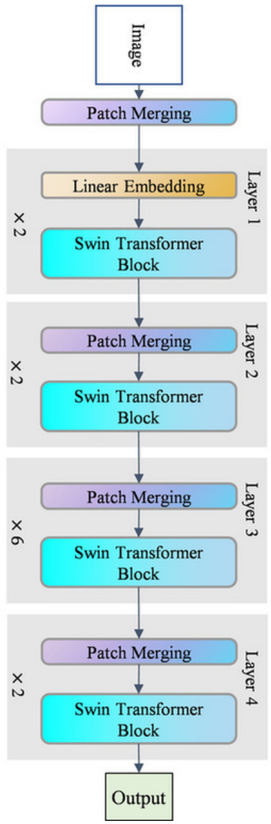
ARCHITECTURAL ABLATION STUDY

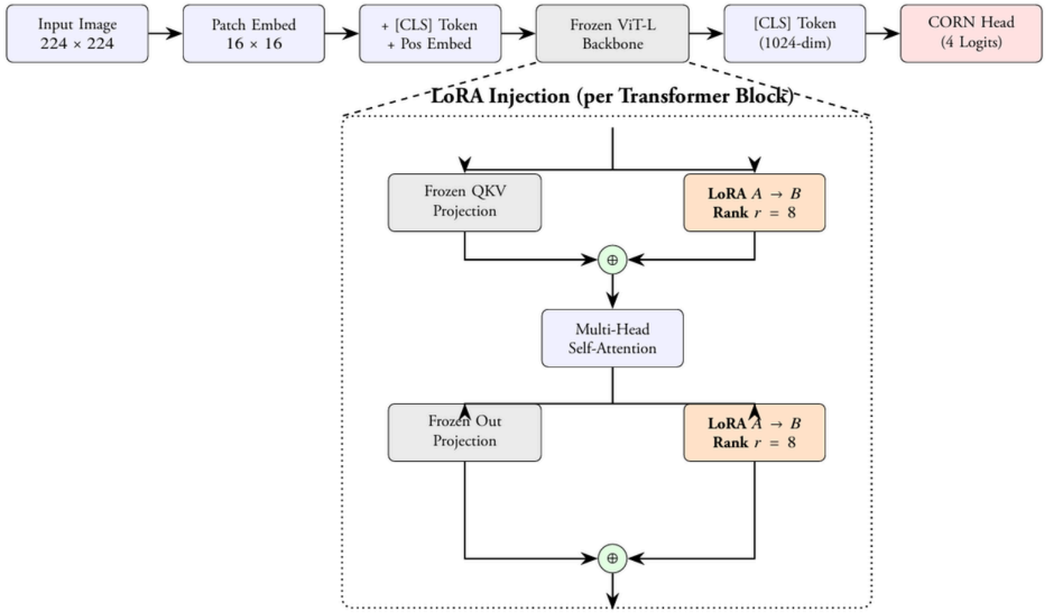
Model A: EfficientNetV2-S. Strong CNN inductive biases (locality). ~21M parameters.

Model B: Swin-V2-Tiny. Hierarchical transformer with shifted-window attention. ~28M parameters.

Model D: RETFound + LoRA. Domain-specific foundation model pre-trained on 1.6M retinal images. Used Low-Rank Adaptation (LoRA)

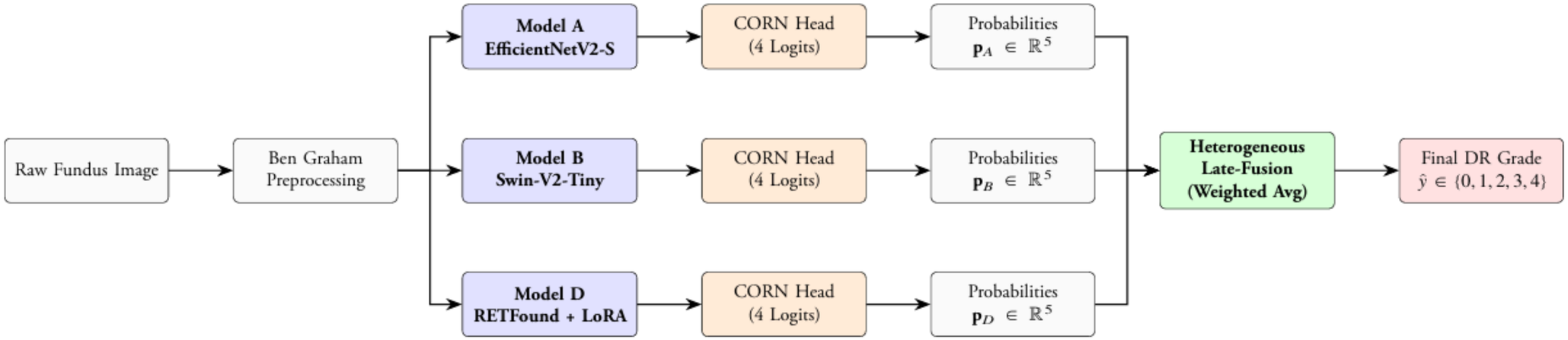






Key Achievement: Fine-tuned RETFound using only ~1.2M trainable parameters (0.4% of total) while freezing the backbone.

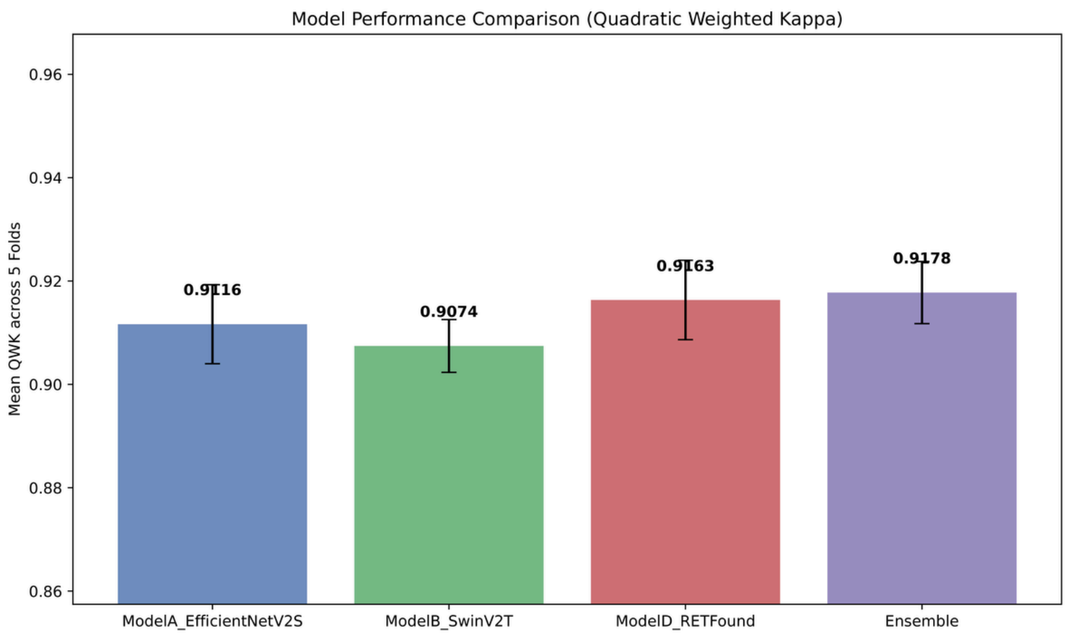
THE END-TO-END ENSEMBLE



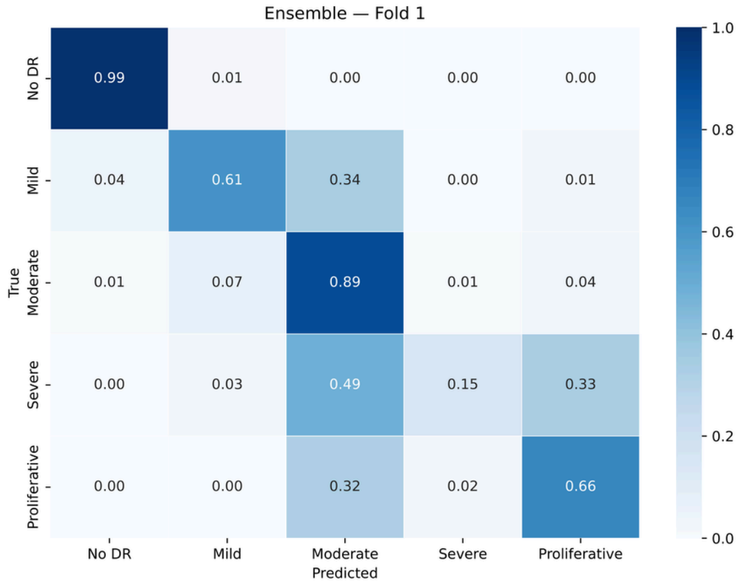
- Heterogeneous late-fusion ensemble.
- Combines class probability distributions from all three models using optimized weighted averaging.

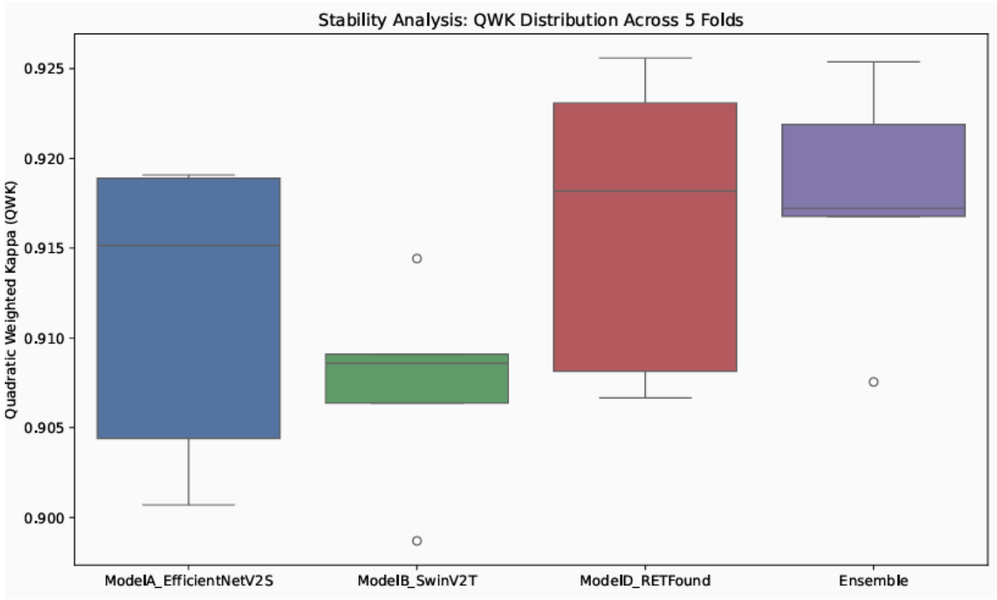
RESULTS – QUANTITATIVE PERFORMANCE

- All models achieved "almost perfect agreement" with clinical annotations (QWK > 0.90).
- RETFound was the best individual model (QWK: 0.9163 ± 0.0077).
- The Ensemble achieved the highest overall score (QWK: 0.9178 ± 0.0060) with improved stability.



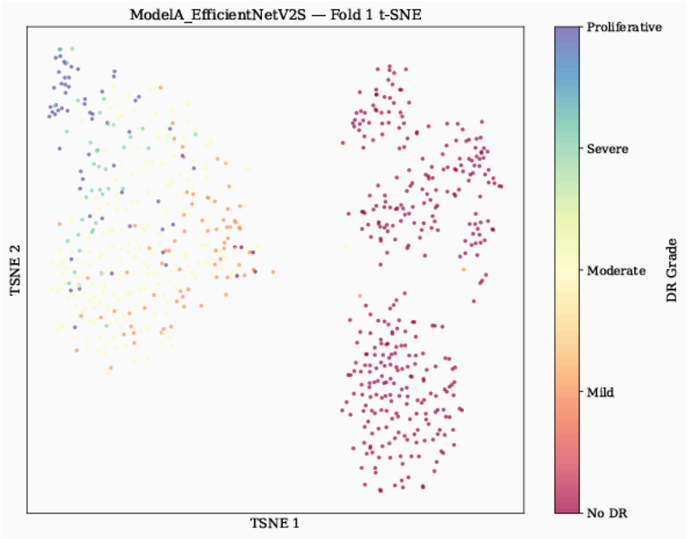
RESULTS – ERROR ANALYSIS & STABILITYCE ANALYSIS (PART 1)



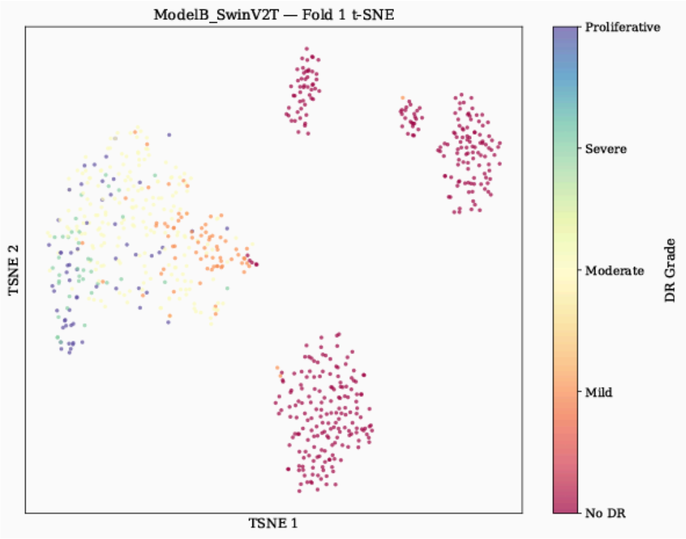


- Misclassifications are predominantly between adjacent grades (e.g., Grade 1 vs. 2), meaning the errors are "clinically safe".
- Distant misclassifications are effectively minimized due to the Ordinal Loss formulation.
- Swin-V2-Tiny was the most stable, but the ensemble brought both high accuracy and low variance.

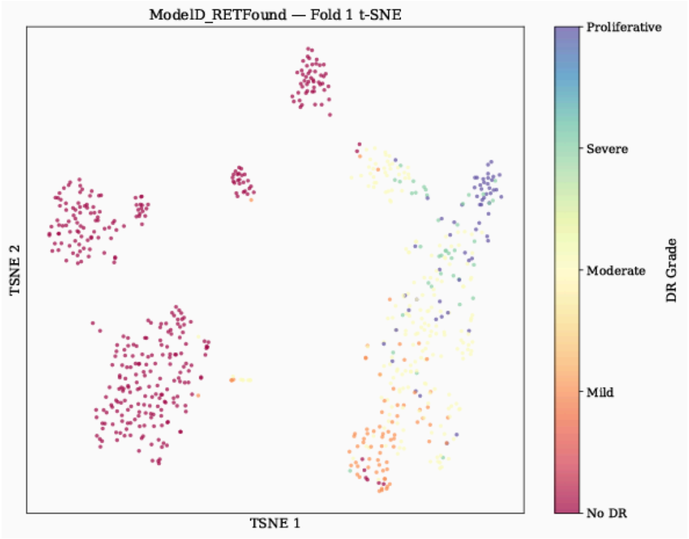
LATENT SPACE VISUALISATION



(a) EfficientNetV2-S



(b) Swin-V2-Tiny

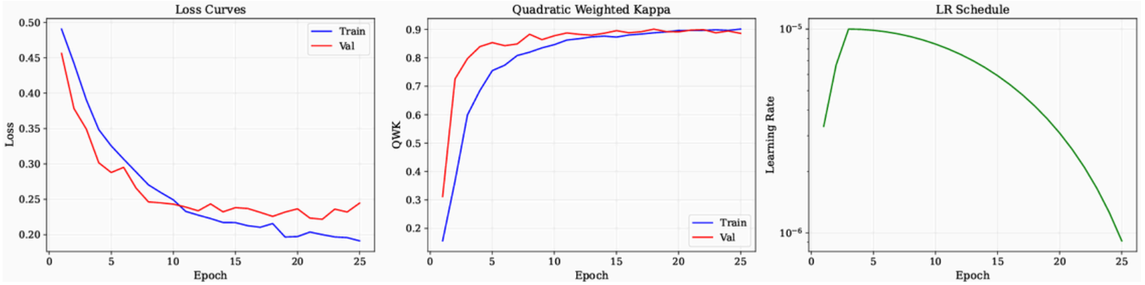


(c) RETFound + LoRA

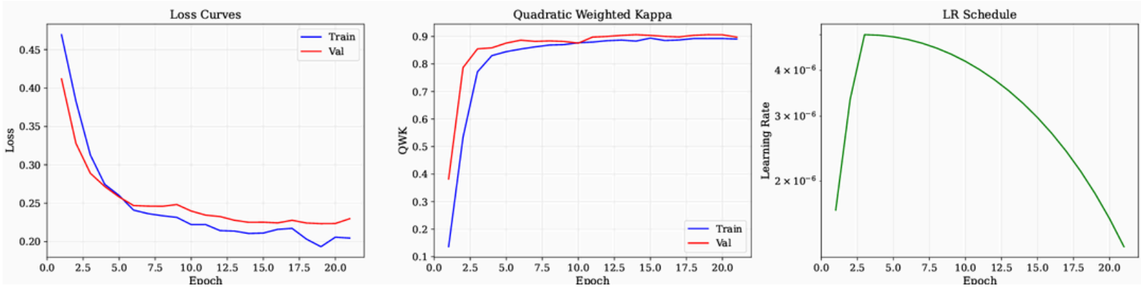
- Visualisations prove the models successfully learned discriminative representations for each severity level.
- The ordinal structure is preserved: adjacent severity grades form neighboring clusters in the latent space.

DISCUSSION & CLINICAL IMPLICATIONS

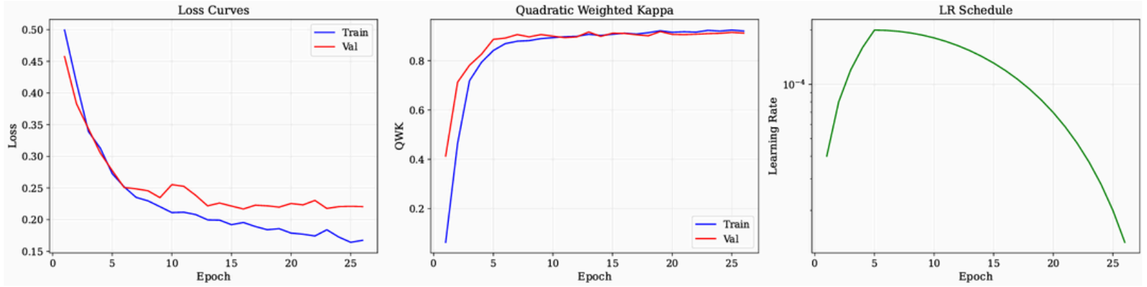
- Foundation Models: Domain-specific pre-training (RETFound) massively outperforms generic ImageNet pre-training, even with 99.6% of parameters frozen.
- CNNs vs. Transformers: CNNs are still highly competitive on smaller datasets due to inductive biases.
- Clinical Impact: A QWK of 0.9178 approaches expert human inter-grader agreement (0.65 - 0.85). This system is highly viable as a triage tool for large-scale screening.



EfficientNetV2-S



Swin-V2-Tiny



RETFound+LoRA

CONCLUSION & FUTURE WORK

- Ordinal regression is fundamentally superior to standard cross-entropy for disease severity tasks.
- Heterogeneous ensembles successfully mitigate individual architecture weaknesses.
- Future Work: Multi-dataset validation (e.g., EyePACS), adding explainability, and testing on larger multi-modal foundation models.

Code Availability: The complete source code, trained models, and documentation for this thesis are publicly available at https://github.com/ayush-145/Thesis.

THANK YOU